

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Architecture

ShopEZ — MERN Stack E-commerce Application

Phase: Project Design Phase

Document Version: 1.1

Date: 11 June 2026

Prepared by: V S S S Manikanta

Classification: Project Documentation — For Academic Submission

Solution Architecture

1. Layered System Architecture

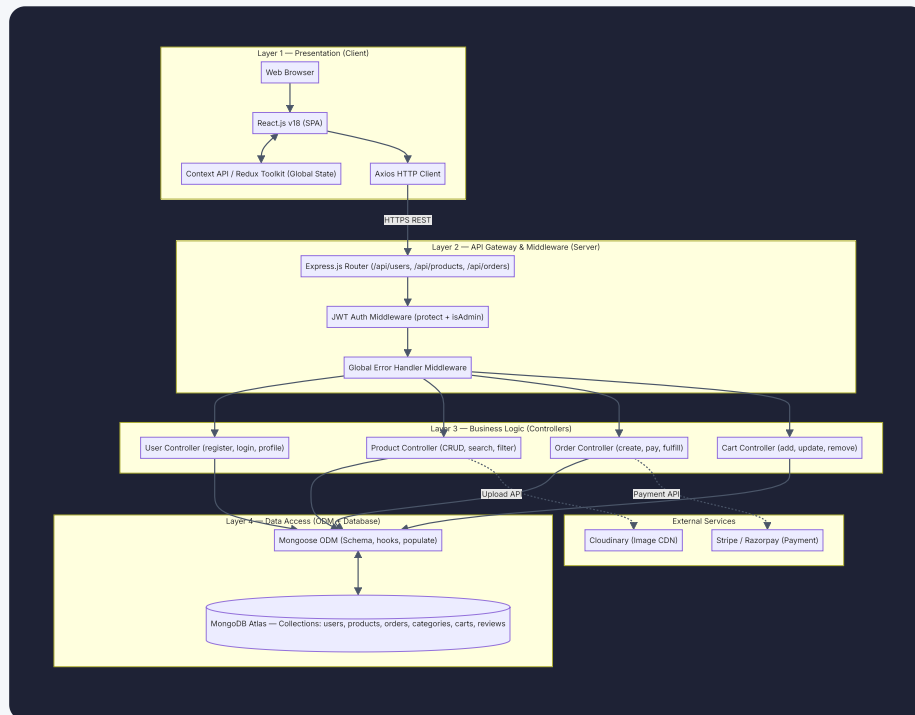


FIGURE 9.1 — 4-LAYER SYSTEM ARCHITECTURE DIAGRAM

2. MVC Pattern Mapping

MVC Layer	MERN Component	Responsibilities
View	React.js (Frontend SPA)	Renders UI components; manages local state; dispatches API calls via Axios; updates global state via Context/Redux
Controller	Express.js Controllers	Receives validated HTTP requests; orchestrates business logic; invokes Model methods; formats and returns JSON responses
Model	Mongoose Schemas & Models	Defines data structure, validation rules, and type constraints; executes lifecycle hooks; interfaces with MongoDB Atlas

3. UML Class Diagram — Mongoose Models



FIGURE 9.2 — UML CLASS DIAGRAM: MONGOOSE DATA MODELS

4. Deployment Topology

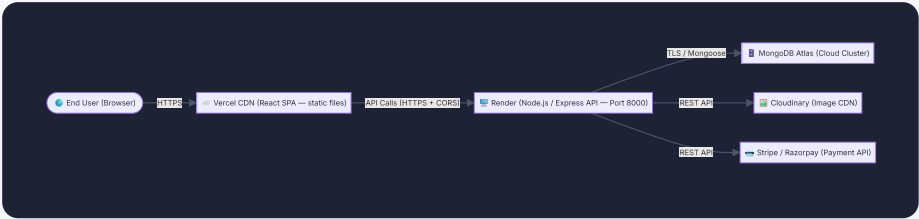


FIGURE 9.3 — CLOUD DEPLOYMENT TOPOLOGY

5. Deployment URL Reference

Component	Platform	URL
Frontend (React SPA)	Vercel	<code>https://e-commerce-application-neon-five.vercel.app/</code>
Backend (Node.js API)	Render	<code>https://shopez-api-c30e.onrender.com</code>
Database	MongoDB Atlas	Cloud-hosted (M0 Free Tier)
Image Storage	Cloudinary	<code>res.cloudinary.com/...</code>